A Project on

CFD ANALYSIS OF ROCKET ENGINE

For the course of Space Flight Mechanics and Propulsion Systems Mech 6251 of Master of Engineering in Aerospace Engineering

Prepared For

Dr. Jerin John

Prepared by

Sri Vyas Narayan Student ID: 40326839

Sanket Bhartiya Student ID: 40321775

DEPARTMENT OF MECHANICAL, INDUSTRIAL & AEROSPACE ENGINEERING (MIAE), GINA CODY SCHOOL OF ENGINEERING AND COMPUTER SCIENCE,

CONCORDIA UNIVERSITY, MONTREAL

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ABSTRACT

Rockets have always fueled human curiosity for more than a thousand years. The preliminary rockets appeared early in China in the forms of fireworks and signaling using gunpowder. They reveled an important idea, Pushing mass out of the object creates propulsive force enabling motion. Centuries later this concept inspired scientists to think about reaching space. In the early 1900's, pioneers like Konstantin Tsiolkovsky and Robert Goddard turned rocketry into real science. In 1926, Goddard launched the first liquid rocket, paving way for further development and innovation in rocketry. During World War II, the Germans created the first human-made object to reach the edge of space, the V-2 rocket. Soon after the U.S and the Soviet Union followed suit and started developing rockets, transforming the sector forever. Many achievements like the Sputnik and Apollo Moon landings followed. These early rockets were single use, meaning they were used once and discarded. This paved the way for the development of reusable rockets in 2010 by SpaceX. They developed the Falcon 9, which had boosters that could land back on the Earth. Reusable suborbital launch vehicles like the X-15 and Blue Origin's New Shepard now fly repeatedly for research and tourism. Taking inspiration from those, with reusability in mind our mission is to design a 1D, 500kN thrust engine which can be used in reusable suborbital launch vehicles.

New Shepard Rocket (Fig 1.1)

INTRODUCTION

The Origin New Shepard rocket is powered by a single BE-3PM engine. It is a high-performance engine which produces almost 500kN of thrust during lift-off. It is a reusable liquid rocket engine specifically designed for vertical take-off and vertical landing (VTVL) missions. The engine uses liquid oxygen as the oxidizer and liquid hydrogen as the fuel. This specific engine is highly efficient as it has the ability to throttle down deeply, to 90kN. This helps the craft to slow the booster to approximately 6 mph which helps in precise landing on the pad. This engine uses a combustion tap-off type engine cycle, where a small amount of hot combustion gases is redirected from the main engine chamber to drive the engine's turbopumps. It also fits in perfectly with the future of aerospace as it is a clean engine, since the only by-products of the hydrogen-oxygen combustion is water vapor without any carbon emissions. The rocket powers the entire ascent has a crew capsule which past the Karman line (100 km), which is also the internationally recognized boundary of space. After the capsule separates, the booster falls back to the Earth. The BE-3PM engines reignite for a controlled descent and vertical landing. The first flight of this spacecraft was on April 29, 2015. As of January 2026, it has successfully completed 38 missions. Despite all the success, Blue Origin has announced a pause in New Shepard tourism flights for at least 2 years to refocus resources on its Blue Moon lunar lander program. There are a few similar engines in its category, like the Merlin 1D which was developed by SpaceX and used in Falcon 9. It is also a reliable and reusable Vertical takeoff and landing (VTVL) engine. But it uses RP-1 instead of hydrogen. The BE-3PM has the advantage in terms of deep throttling down when compared with Merlin 1D engine. There are various upper stage engines like RL10 and J-2S engines but they are not designed for sea level conditions and reusability.

METHODOLOGY

The entire project is done using a calculator and 4 different softwares. Primarily, the constant parameters were identified such as the thrust requirement and used to calculate all the other unknown parameters. These parameters varies based on the mission requirements and are a key pillar in the calculation of the geometry of the engine model.

CEA (NASA) is an online calculator used for rocket engine calculations for combustion and flow of various fuels, one of which was gaseous hydrogen as fuel and gaseous oxygen as oxidizer. CEA produced the properties such as temperature of combustion chamber, the characteristic exhaust velocity, the effective exhaust velocity and many more. A MATLAB code was developed which is designed to take user input for the critical values such as ideal thrust, pressure of chamber and the O/F ratio to calculate the isp and temperatures. With all those values the radius and length of the combustion chamber, throat and exit were calculated.

The calculated values were used to design a 2D model in SOLIDWORKS. The 2D sketch was converted to a surface and exported as .STEP file to be imported into Ansys. Ansys Fluent was the software used to simulate the condition of the engine. The model was modified further to optimize for better performance.

3.1 Parameters

The parameters were determined using a few constants. Those constants are,

Thrust = 500kN

Chamber pressure = 1000 Psia

= 6894757.28 Pa

Characteristic exhaust velocity (C*) = 3998.1 m/s

This was obtained from CEA for our operating conditions.

Oxidizer to Fuel Ratio = 3.2

The O/F ratio was chosen based on the results from the CEA and from literature review. We did a range for the calculation of O/F and found this gave us the highest specific impulse.

We also have $P_2 = P_3$ as the engine is operating optimally and since it is at sea level, the outlet pressure is 101325 Pa. All the density values for hydrogen and oxygen were obtained from Engineering Toolbox.

The specific impulse (Isp) = 407.5 s

Using the O/F ratio,

The mass flow rate of oxygen was calculated to be 95.28 Kg/s and the mass flow rate of hydrogen was calculated to be 29.779 Kg/s.

Temperature

Then we proceeded to use the obatined values to calculate the mass flow rate,

F =m*v

Therefore we got,

Mass flow rate = 125.059 kg/s

The temperature was calculated using the different gamma values and compared with CEA values.

$T_t = T_c \times 2 / (\gamma + 1)$

$T_t = 3178.49$ K

Similarly Temperature at exit was calculated to be

$T_e = 1676.2$ K

3.2 Geometry

$\dot{m} = \rho A V$

Here $\rho = 19.84$ Kg/s

$\dot{m} = 125.059$ kg/s

V = 1743 m/s

Therefore the R_t was calculated to be 60.689 mm.

The contraction ratio was 5.2925, the characteristic length of the combustion chamber was found to be 56 cm from literature, and therefore the combustion chamber radius was calculated to be 81.89 mm.

The expansion ratio was 7.8489 and therefore the exit radius was calculated to be 170.02 mm.

From the above mass flow rate of hydrogen and oxygen, the total area required for their injection was calculated as,

$A_h = 0.00891\ m^2$

$A_o = 0.01077\ m^2$

The total area was divided into multiple sections to allow for better injection and mixing.

The values obtained above was compared to the data from CEA and the values from MATLAB and was found to be within 7% error for the highest difference.

```
            THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

          COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin =  1000.0 PSIA
CASE = _______________

            REACTANT                    WT FRACTION      ENERGY      TEMP
                                         (SEE NOTE)     KJ/KG-MOL      K
FUEL        H2                           1.0000000        0.000      0.000
OXIDANT     O2                           1.0000000        0.000      0.000

O/F=    3.20000  %FUEL= 23.809524  R,EQ.RATIO= 2.480213  PHI,EQ.RATIO= 2.480213

                CHAMBER   THROAT     EXIT
Pinf/P            1.0000   1.7850   68.027
P, BAR            68.947   38.626   1.0135
T, K             2796.59  2531.37  1197.87
RHO, KG/CU M    2.4991 0 1.5506 0 8.6159-2
H, KJ/KG         0.00000 -1520.30 -7992.58
U, KJ/KG        -2758.84 -4011.35 -9168.92
G, KJ/KG        -67686.0 -62787.2 -36984.7
S, KJ/(KG)(K)    24.2030  24.2030  24.2030

M, (1/n)           8.428    8.449    8.467
(dLV/dLP)t      -1.00228 -1.00104 -1.00000
(dLV/dLT)p        1.0468   1.0234   1.0000
Cp, KJ/(KG)(K)    6.2098   5.6698   4.2699
GAMMAs            1.2074   1.2206   1.2987
SON VEL,M/SEC     1825.1   1743.7   1236.0
MACH NUMBER        0.000    1.000    3.235

PERFORMANCE PARAMETERS

Ae/At                     1.00000   7.8489
CSTAR, M/SEC               2550.0   2550.0
CF                         0.6838   1.5679
Ivac, M/SEC                3172.3   4292.4
Isp, M/SEC                 1743.7   3998.1


MASS FRACTIONS

*H               0.00090  0.00043  0.00000
*H2              0.14139  0.14174  0.14210
H2O              0.85455  0.85668  0.85790
*O               0.00002  0.00000  0.00000
*OH              0.00313  0.00115  0.00000
*O2              0.00001  0.00000  0.00000

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
```

CEA Analysis (Fig 3.21)

3.3 Propellants

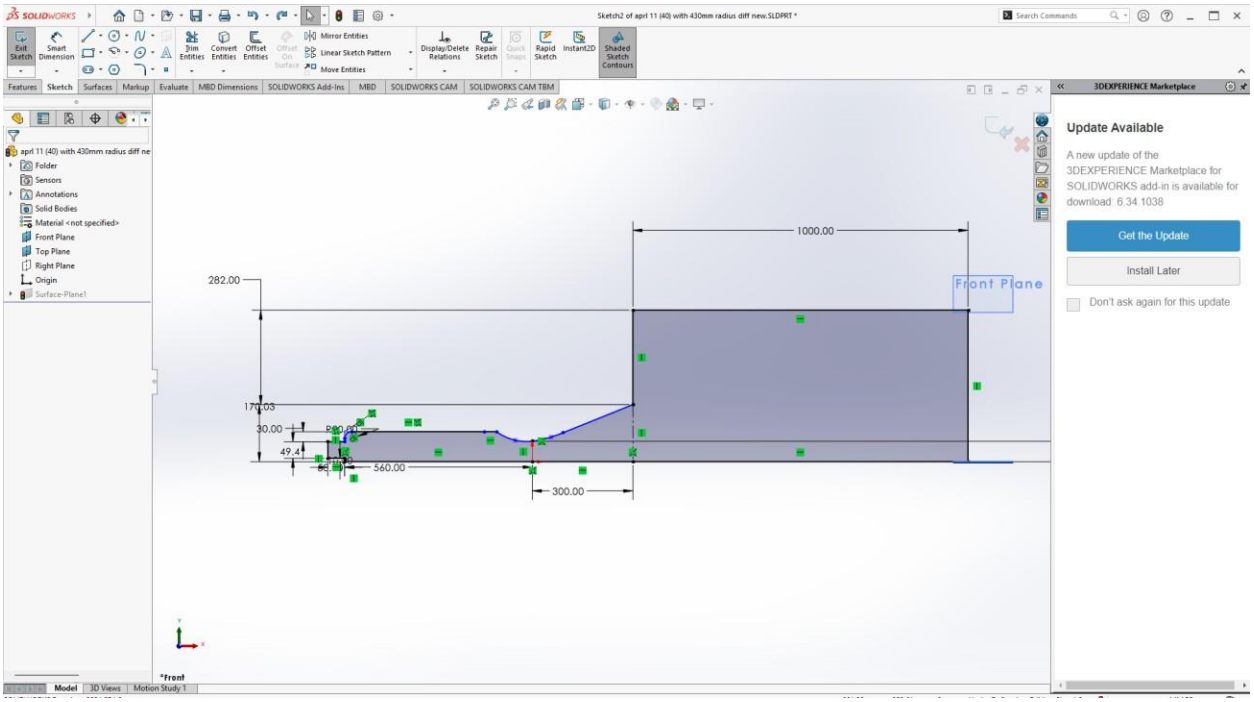
We decided to go with hydrogen as fuel and oxygen as oxidizer since it was used in the New Shepard rocket's BE-3PM engine. It produces no harmful by-products and the combustion is "clean", which is known to be the future of aerospace and aviation. This engine also produces high specific impulse and thrust which is used to take people to the Karman line. Using methane as fuel and oxygen as oxidizer such as the ones used by SpaceX (Raptor engine) was also a thought for a while but decided not to as the efficiency of those engines are lower than the BE-3PM engine. Another reason was Blue Origin is upgrading the BE-3PM engine and enabling it to operate as an upper stage engine in their upcoming rocket New-Glenn, thereby enabling us to work on this project further than just the one engine we decided on.

SIMULATION

The simulation as mentioned earlier was done in Ansys while the design was modeled in SolidWorks.

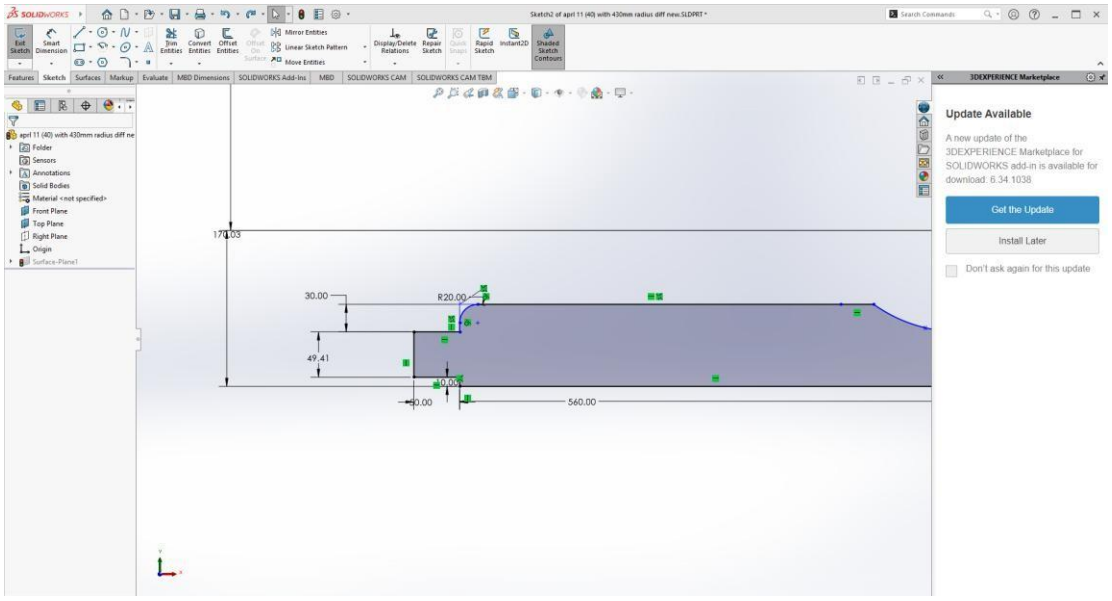
4.1 Design

The geometry was designed in SolidWorks with the above mentioned areas for the injector, combustion chamber, and nozzle. A large area back box was added to visualize the flow coming out of the nozzle better.



Geometry (Fig 4.11)

The length of the back box was chosen based on review. This model was sketched on XY Plane with the throat being at the Y axis so, the nozzle exit was at 300 mm in x axis. The nozzle geometry was drawn using spline tool. This sketch was then converted into a surface and translated to fluent for further analysis.

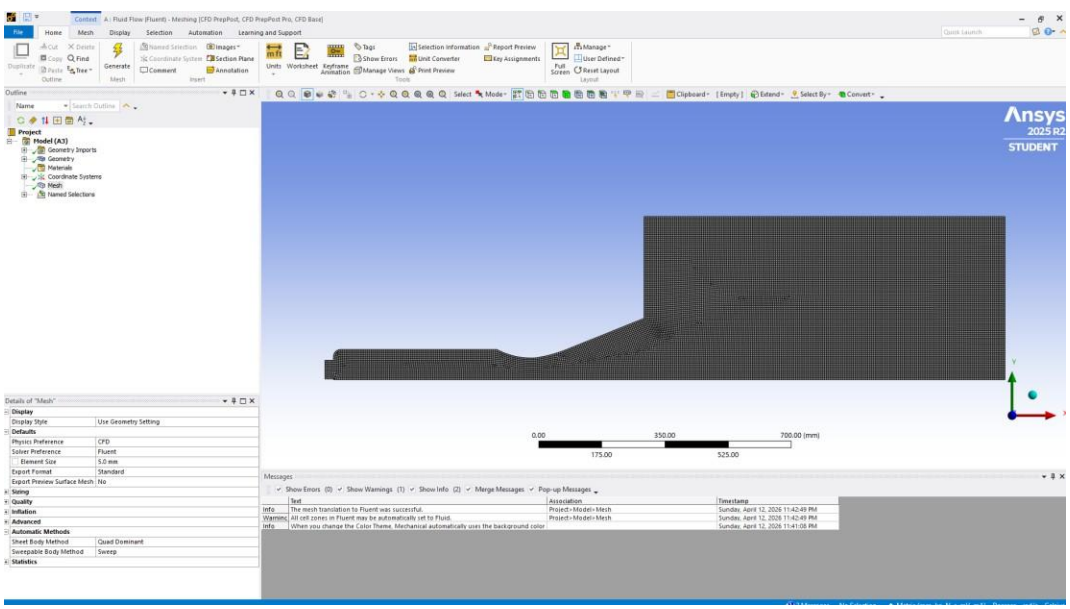


Geometry (Fig 4.12)

A reimagined 2D version of the pintle type injector, in this case a coaxial annular type injector was chosen as the air is fed into the injector section axially while the engine where as fuel was injected radially. This helps better in the mixing of the fuel and oxidizer since the fuel flowing radially is going to be at a higher velocity than the oxidizer flowing into the injector axially.

4.2 Mesh

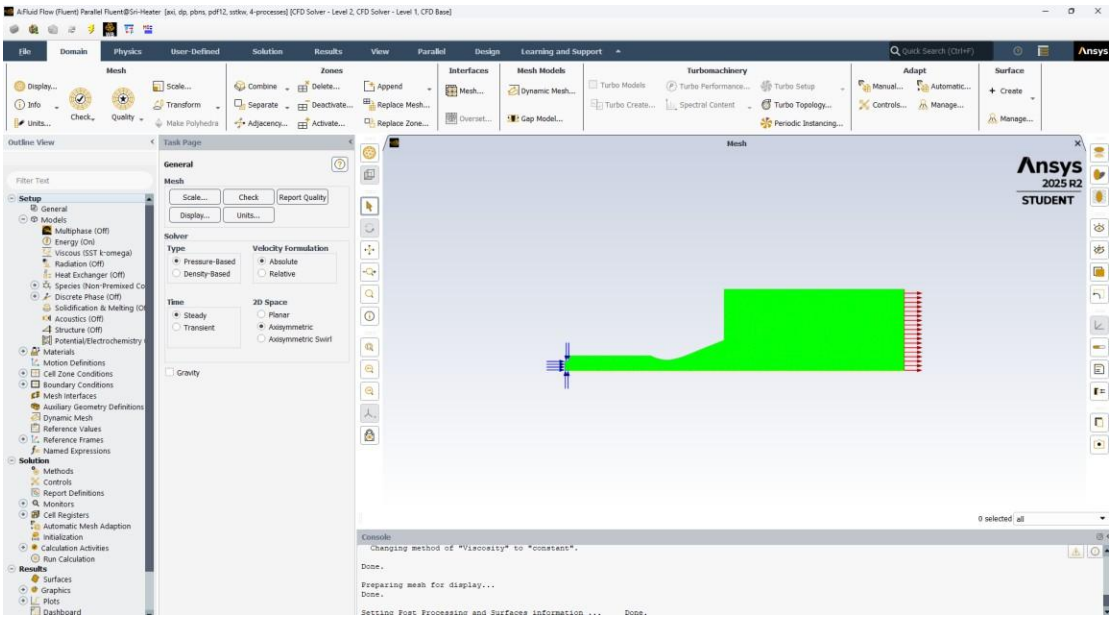
The model was meshed finely with good quality. The element nodes were 5 mm in size.



Meshing (Fig 4.21)

4.3 Setup

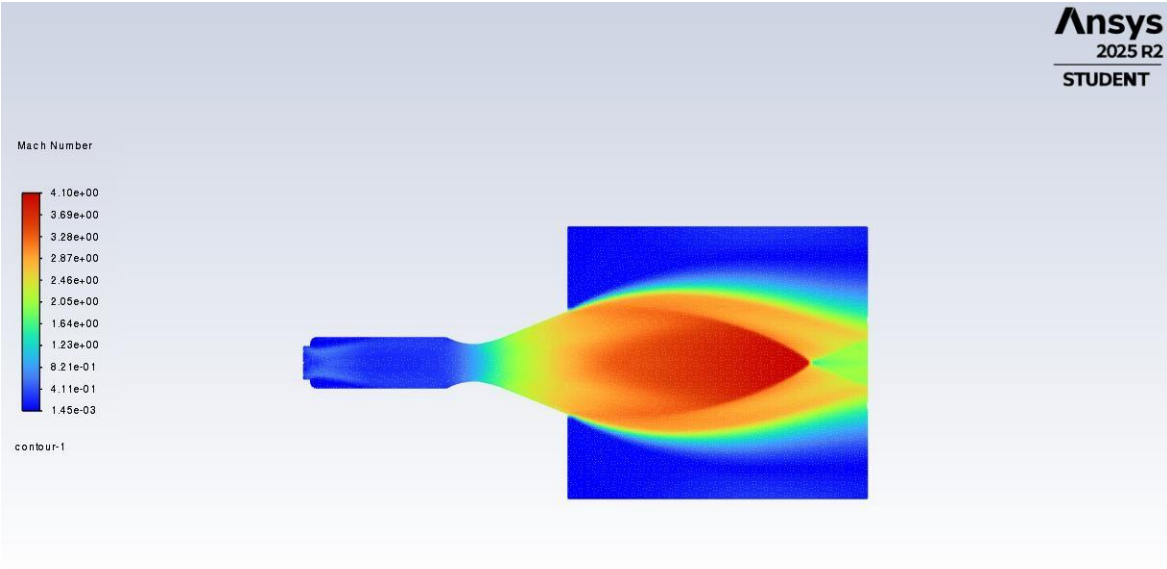
The model was loaded into fluent and the conditions were given. Energy equation was turned on, the turbulence model used was k-omega SST. Non-premixed combustion was used with compressibility effects turned on, Correct chamber pressure was given and the table was generated. Boundary conditions were specified based on the calculations and the model used hybrid initialization. The model was run for 10000 iterations with convergence turned off.



Fluent (Fig 4.31)

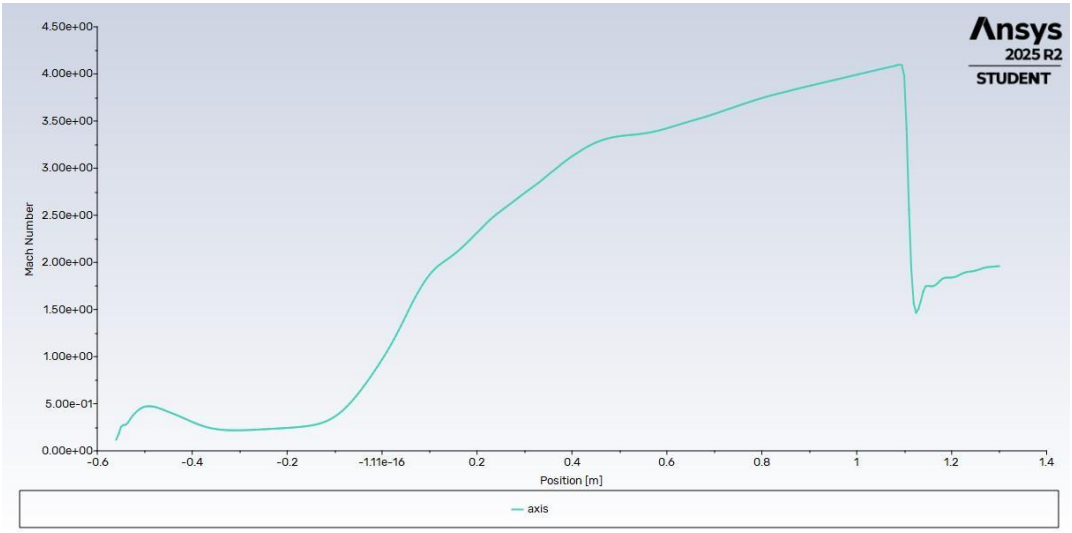
RESULTS

After running the model, the following results were observed



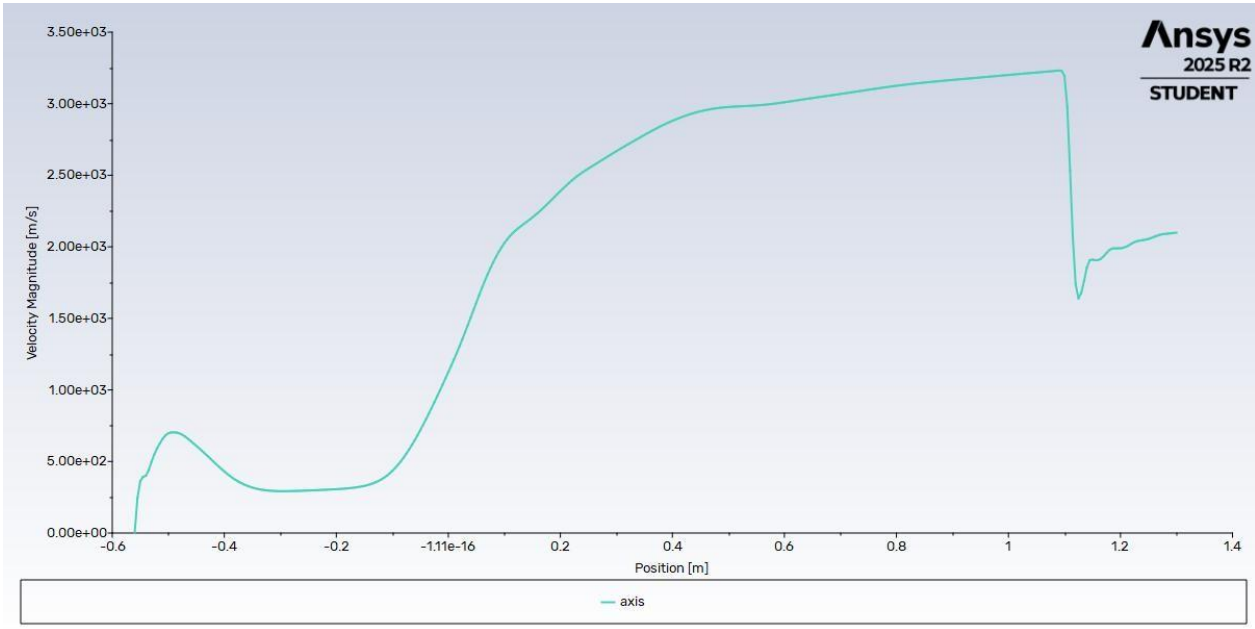
Mach number contour (Fig 5.1)

From the Mach number contour, We can see the flow starts very low and subsonic at the entry of hydrogen and oxygen, the flow reaches almost stagnation velocity at the exit of the nozzle when compared to the velocity downstream. Then the flow forms a diamond shape displaying the features of a properly expanded nozzle. We can see that the Mach number at throat is 1, which is the ideal condition to get proper expansion from a converging diverging nozzle and this can be seen more clearly in Fig 5.2.



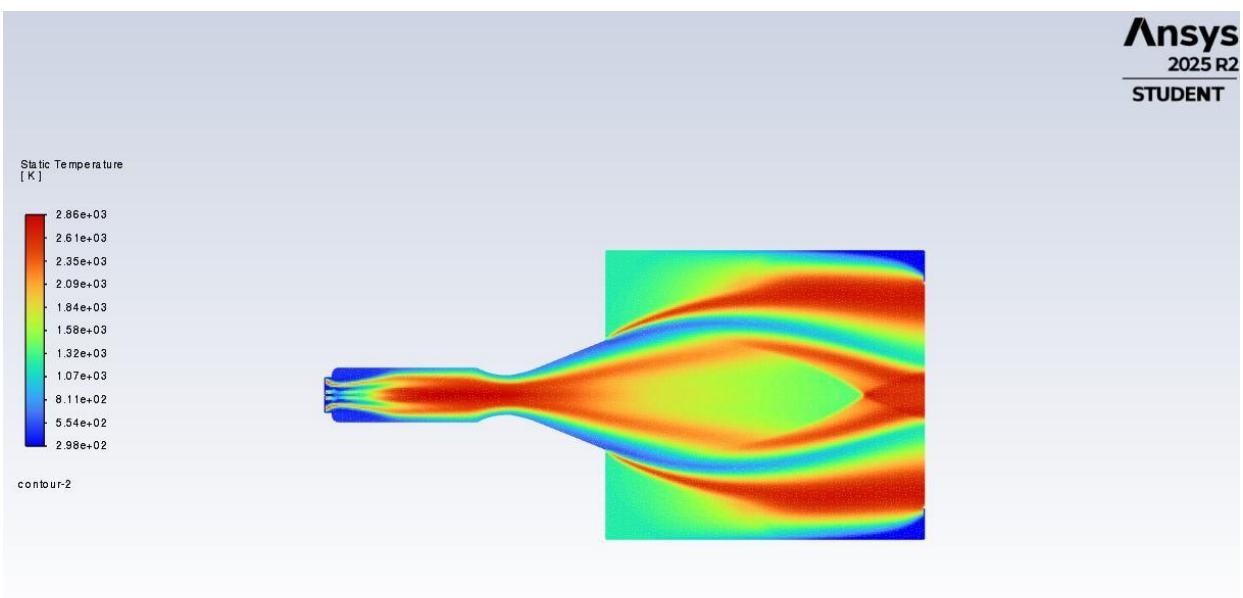
Mach Number Plot (Fig 5.2)

As mentioned earlier 0.3m is the exit of the nozzle in Fig 5.2. The velocity of the air was affected by the temperature and the medium of translation.

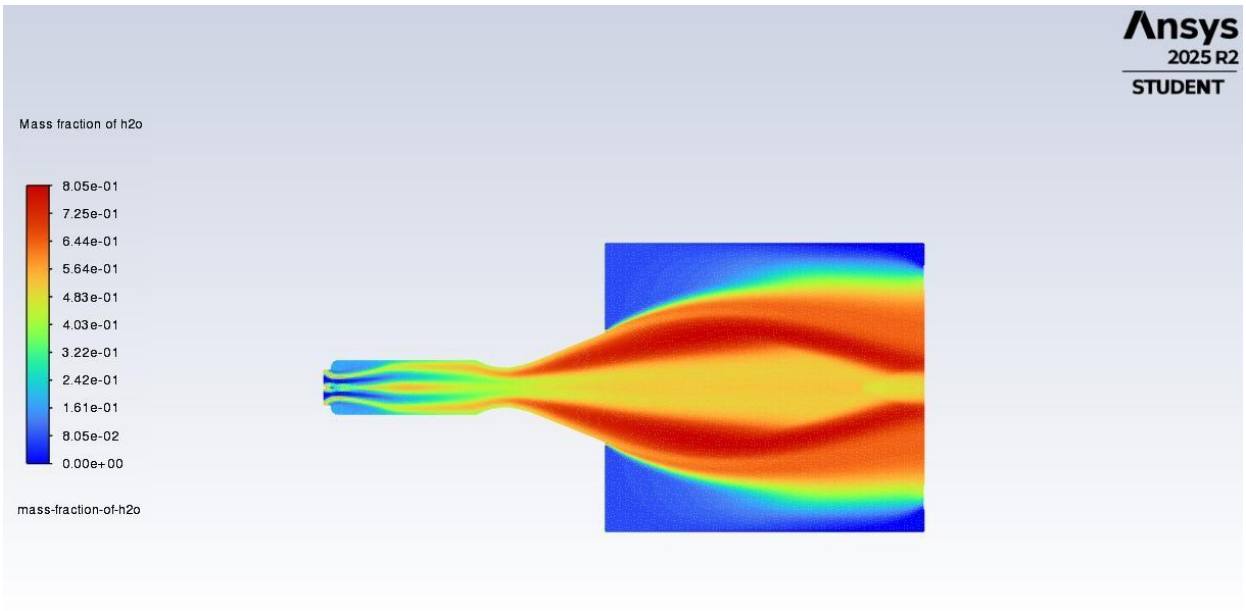


Velocity Plot (Fig 5.3)

From the velocity plot we can see that the velocity at starts at around 400 m/s in the pre chamber, the small peak may be due to high speed hydrogen injection. It dips subsequently and settles before rapidly rising in the nozzle section of the geometry, reaching over 3200 m/s at its peak outside the nozzle. At the exit of the nozzle the velocity is close to 2650 m/s at 0.3m in the graph.

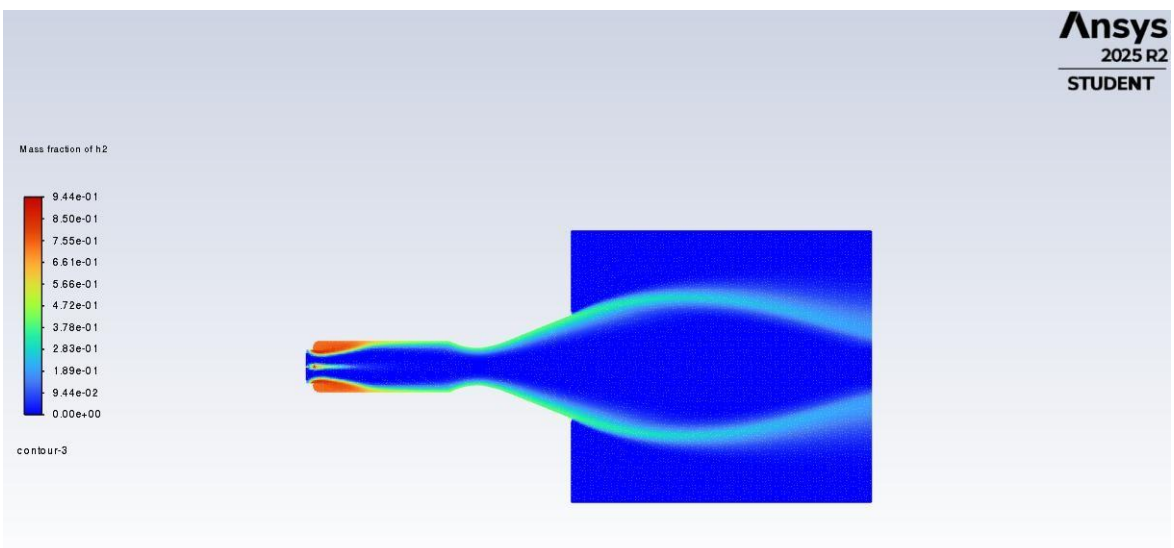


Temperature plot (Fig 5.4)



Mass fraction of H_2O (Fig 5.5)

The mass fraction of H_2O clearly shows the streamlines where the mixture fraction is highest, and the other parts maybe occupied by non-mixed hydrogen since the case taken for this simulation is fuel rich.



Mass fraction of H_2 (Fig 5.6)

This contour shows the distribution of hydrogen inside the nozzle. The deep red region near the nozzle indicates a very high concentration of hydrogen, but downstream, the jet expands into the combustion chamber, and the concentration drops to green and finally to blue which indicates very little to no hydrogen.

CONCLUSION

The simulation was done and the results obtained almost lines up with the calculations done before and the exit velocity was found to be higher. CEA proved to be a useful tool that explained the range of the values and aligned with the manual calculations and the MATLAB calculations. But it did not correspond exactly to the velocity found in the simulations. The error found between our calculations and CEA can be because the CEA was done as an average value for a 3D model with losses (as CEA calculates the thermodynamic losses from frozen chemistry) whereas the calculations done by us did not account for those losses and gave us the theoretical performance. The values from RPA did account for the losses but was found nowhere to be close to either CEA or our calculations.

The calculations and simulations show a theoretical possibility of higher thrust (almost 520 kN) from the original New Shepard rocket (500 kN), outperforming it in a 2D simulation. This maybe because the values taken for the calculation of the thrust from the simulation were not average of the entire surface but rather the value at the line of the axis, and the reason for higher thrust can be mitigated by calculating for the losses in the real world.

Future work on this project can include a better bell nozzle which can reduce the mass of the rocket and increase efficiency of the engine at the same time and adding regenerative cooling channel along the walls to reduce the temperature. The work can also dive further into the design of the injector to improve mixing, which is a major component for combustion which in turn affects the thrust of the entire rocket.

REFERENCES

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2. **Modern Engineering for Design of Liquid-Propellant Rocket Engines**
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4. http://www.braeunig.us/space/propuls.htm
5. https://www.engineeringtoolbox.com/
6. https://2025.help.altair.com/2025/flowsim/topics/internal/solver_technical_manual/elements/elements_combustion_element_internal.htm#:~:text=Combustion%20Element%20General%20Description,Available%20loss%20options%20are:

APPENDIX

The following is the MATLAB code for the calculation of the parameters.

```
clc; clear; close all;
%% ── USER INPUTS ──────────────────────────────────────
F = input('Enter required thrust (N): ');
P_c = input('Enter chamber pressure (Pa): ');
OF = input('Enter oxygen to fuel ratio: ');
% Fixed conditions
P_exit = 101325; % Exit (ambient) pressure [Pa] — sea level
g0 = 9.80665; % Standard gravity [m/s²]
R_univ = 8314; % Universal gas constant [J/(kmol·K)]
cea_table = [
1.0, 2972.4, 1.167, 7.24;
1.5, 3116.9, 1.175, 8.18;
2.0, 3243.8, 1.182, 9.11;
2.5, 3397.1, 1.196, 10.12;
3.0, 3479.6, 1.207, 11.14;
3.2, 3517.0, 1.213, 11.82;
3.5, 3545.2, 1.219, 12.20;
4.0, 3563.6, 1.228, 13.27;
4.5, 3531.2, 1.237, 14.24;
5.0, 3474.8, 1.247, 15.17;
5.5, 3399.5, 1.257, 16.01;
6.0, 3310.7, 1.265, 16.75;
6.5, 3211.5, 1.273, 17.40;
7.0, 3104.3, 1.280, 17.96;
7.5, 2991.8, 1.286, 18.44;
8.0, 2875.9, 1.292, 18.86;
9.0, 2636.8, 1.301, 19.54;
10.0, 2397.2, 1.309, 20.08;
];
% Clamp O/F to table range
OF_clamped = max(cea_table(1,1), min(cea_table(end,1), OF));
if OF < cea_table(1,1) || OF > cea_table(end,1)
warning('O/F = %.2f is outside table range [%.1f, %.1f]. Clamped.', ...
OF, cea_table(1,1), cea_table(end,1));
end
% Interpolate all three properties
T_c = interp1(cea_table(:,1), cea_table(:,2), OF_clamped, 'pchip');
gamma = interp1(cea_table(:,1), cea_table(:,3), OF_clamped, 'pchip');
M_mol = interp1(cea_table(:,1), cea_table(:,4), OF_clamped, 'pchip');
% Derived gas properties
R_gas = R_univ / M_mol; % Specific gas constant [J/(kg·K)]
Cp = gamma * R_gas / (gamma - 1); % Specific heat [J/(kg·K)]
fprintf('\n--- INPUT PARAMETERS ---\n');
```

```
fprintf(' Thrust F = %.0f N\n', F);
fprintf(' Chamber pressure P_c = %.3f MPa\n', P_c/1e6);
fprintf(' O/F ratio = %.2f\n', OF);
fprintf(' Exit pressure P_e = %.0f Pa\n', P_exit);
fprintf('\n--- CEA COMBUSTION GAS PROPERTIES (interpolated) ---\n');
fprintf(' Flame temperature T_c = %.2f K\n', T_c);
fprintf(' Gamma γ = %.4f\n', gamma);
fprintf(' Mol. weight M_mol = %.3f kg/kmol\n',M_mol);
fprintf(' Gas constant R = %.3f J/(kg·K)\n',R_gas);
fprintf(' Cp = %.3f J/(kg·K)\n',Cp);
%% ── THROAT CONDITIONS (M = 1) ──────────────────────────────
M_t = 1.0;
T_t = T_c * (2 / (gamma + 1));
P_t = P_c * (2 / (gamma + 1))^(gamma / (gamma - 1));
rho_t = P_t / (R_gas * T_t);
a_t = sqrt(gamma * R_gas * T_t);
V_t = a_t; % velocity at throat = speed of sound
fprintf('\n--- THROAT CONDITIONS ---\n');
fprintf(' Mach number M_t = %.4f\n', M_t);
fprintf(' Temperature T_t = %.2f K\n', T_t);
fprintf(' Pressure P_t = %.4f MPa\n',P_t/1e6);
fprintf(' Speed of sound a_t = %.2f m/s\n',a_t);
fprintf(' Density rho_t = %.4f kg/m³\n',rho_t);
%% ── EXIT MACH NUMBER ──────────────────────────────────────
% Solve isentropic pressure ratio for M_e (supersonic root)
pressure_ratio = P_c / P_exit;
f_mach = @(M) (1 + (gamma-1)/2 * M^2)^(gamma/(gamma-1)) - pressure_ratio;
options = optimset('Display','off','TolX',1e-12,'TolFun',1e-12);
M_e = fzero(f_mach, 4.0, options);
T_e = T_c / (1 + (gamma-1)/2 * M_e^2);
P_e = P_c / (1 + (gamma-1)/2 * M_e^2)^(gamma/(gamma-1));
rho_e = P_e / (R_gas * T_e);
a_e = sqrt(gamma * R_gas * T_e);
V_e = M_e * a_e;
fprintf('\n--- EXIT CONDITIONS ---\n');
fprintf(' Mach number M_e = %.4f\n', M_e);
fprintf(' Temperature T_e = %.2f K\n', T_e);
fprintf(' Pressure P_e = %.2f Pa\n', P_e);
fprintf(' Exit velocity V_e = %.2f m/s\n', V_e);
fprintf(' Density rho_e = %.6f kg/m³\n',rho_e);
C_star = (1/gamma) * sqrt(gamma * R_gas * T_c) * ...
((gamma+1)/2)^((gamma+1)/(2*(gamma-1)));
% Ideal thrust coefficient (perfectly expanded, P_e = P_amb)
term1 = (2*gamma^2/(gamma-1)) * (2/(gamma+1))^((gamma+1)/(gamma-1));
term2 = 1 - (P_exit/P_c)^((gamma-1)/gamma);
C_F = sqrt(term1 * term2);
% Note: pressure thrust = 0 when perfectly expanded
```

```
% Effective exhaust velocity
Ve_eff = C_F * C_star;
% Specific impulse
Isp = Ve_eff / g0;
fprintf('\n--- EXHAUST VELOCITY & PERFORMANCE ---\n');
fprintf(' Char. velocity C* = %.2f m/s\n', C_star);
fprintf(' Thrust coefficient C_F = %.4f\n', C_F);
fprintf(' Eff. exhaust vel. Ve = %.2f m/s\n', Ve_eff);
fprintf(' Specific impulse Isp = %.2f s\n', Isp);
m_dot = F / Ve_eff;
m_dot_ox = m_dot * OF / (1 + OF);
m_dot_fuel = m_dot / (1 + OF);
fprintf('\n--- MASS FLOW RATES ---\n');
fprintf(' Total mdot = %.4f kg/s\n', m_dot);
fprintf(' LOX (O2) mdot_ox = %.4f kg/s\n', m_dot_ox);
fprintf(' LH2 (H2) mdot_fuel = %.4f kg/s\n', m_dot_fuel);
% Verify thrust
F_check = m_dot * Ve_eff;
fprintf(' Thrust check F = mdot x Ve = %.2f N\n', F_check);
%% ── THROAT AND EXIT AREAS ──────────────────────────────
A_t = m_dot / (rho_t * V_t);
D_t = 2 * sqrt(A_t / pi);
% Area-Mach relation at exit
A_ratio_exit = (1/M_e) * ((2/(gamma+1)) * ...
(1 + (gamma-1)/2 * M_e^2))^((gamma+1)/(2*(gamma-1)));
A_e = A_ratio_exit * A_t;
D_e = 2 * sqrt(A_e / pi);
fprintf('\n--- NOZZLE AREA RATIOS ---\n');
fprintf(' Throat area A_t = %.6f m²\n', A_t);
fprintf(' Throat diameter D_t = %.4f m (%.1f mm)\n', D_t, D_t*1000);
fprintf(' Exit/throat ratio eps = %.4f\n', A_ratio_exit);
fprintf(' Exit area A_e = %.6f m²\n', A_e);
fprintf(' Exit diameter D_e = %.4f m (%.1f mm)\n', D_e, D_e*1000);
%% ── CHAMBER GEOMETRY ────────────────────────────────────
contraction_ratio = 4.0;
A_c = contraction_ratio * A_t;
D_c = 2 * sqrt(A_c / pi);
L_star = 1.07; % characteristic length H2/O2 [m]
L_c = L_star * A_t / A_c;
L_conv = (sqrt(A_c/pi) - sqrt(A_t/pi)) / tan(30*pi/180);
L_n = 0.8 * (sqrt(A_e/pi) - sqrt(A_t/pi)) / tan(15*pi/180);
L_total = L_c + L_conv + L_n;
fprintf('\n--- CHAMBER GEOMETRY ---\n');
fprintf(' Contraction ratio Ac/At = %.2f\n', contraction_ratio);
fprintf(' Chamber diameter D_c = %.4f m (%.1f mm)\n', D_c, D_c*1000);
fprintf(' Chamber area A_c = %.6f m²\n', A_c);
fprintf(' Chamber length L_c = %.4f m (%.1f mm)\n', L_c, L_c*1000);
```

```
fprintf(' Convergence length L_cv = %.4f m (%.1f mm)\n', L_conv, L_conv*1000);
fprintf(' Nozzle length L_n = %.4f m (%.1f mm)\n', L_n, L_n*1000);
fprintf(' Total engine length L_tot = %.4f m (%.1f mm)\n', L_total, L_total*1000);
%% ── INJECTOR INLET SIZING (5 inlets, 11 MPa) ──────────────
P_inj = 11e6;
T_inj = 300;
n_inlets = 5;
M_O2 = 0.032;
M_H2 = 0.002;
rho_O2 = P_inj * M_O2 / (R_univ/1000 * T_inj);
rho_H2 = P_inj * M_H2 / (R_univ/1000 * T_inj);
v_O2 = 20;
v_H2 = 30;
A_O2_per = (m_dot_ox / n_inlets) / (rho_O2 * v_O2);
A_H2_per = (m_dot_fuel / n_inlets) / (rho_H2 * v_H2);
D_O2_per = 2 * sqrt(A_O2_per / pi);
D_H2_per = 2 * sqrt(A_H2_per / pi);
fprintf('\n--- INJECTOR INLET SIZING (5 inlets each, 11 MPa, 300 K) ---\n');
fprintf(' LOX inlet diameter (each) = %.4f m (%.1f mm)\n', D_O2_per, D_O2_per*1000);
fprintf(' LOX slot height (each) = %.1f mm | velocity = %.1f m/s\n', D_O2_per*1000,
v_O2);
fprintf(' LH2 inlet diameter (each) = %.4f m (%.1f mm)\n', D_H2_per, D_H2_per*1000);
fprintf(' LH2 slot height (each) = %.1f mm | velocity = %.1f m/s\n', D_H2_per*1000,
v_H2);
%% ── MACH / T / P DISTRIBUTIONS FOR PLOTTING ───────────────
M_sub = linspace(0.01, 1.0, 300);
M_sup = linspace(1.0, M_e, 400);
AR_fn = @(M) (1./M) .* ((2/(gamma+1)) .* ...
(1 + (gamma-1)/2 .* M.^2)).^((gamma+1)/(2*(gamma-1)));
T_fn = @(M) T_c ./ (1 + (gamma-1)/2 .* M.^2);
P_fn = @(M) P_c ./ (1 + (gamma-1)/2 .* M.^2).^(gamma/(gamma-1));
AR_sub = AR_fn(M_sub); T_sub = T_fn(M_sub); P_sub = P_fn(M_sub);
AR_sup = AR_fn(M_sup); T_sup = T_fn(M_sup); P_sup = P_fn(M_sup);
%% ── PLOTS ──────────────────────────────────────────────────
figure('Name','Rocket Engine Design v2', ...
'Position',[80 80 1150 820], 'Color','white');
% 1 — Mach vs Area ratio
subplot(2,3,1);
plot(AR_sub, M_sub, 'b-', 'LineWidth',2); hold on;
plot(AR_sup, M_sup, 'r-', 'LineWidth',2);
plot(1.0, 1.0, 'ko', 'MarkerFaceColor','k', 'MarkerSize',8);
plot(A_ratio_exit, M_e, 'rs', 'MarkerFaceColor','r', 'MarkerSize',8);
xlabel('Area ratio A/A_t [-]'); ylabel('Mach number M [-]');
title('Mach number vs Area ratio');
legend('Subsonic','Supersonic','Throat M=1', ...
sprintf('Exit M=%.2f',M_e),'Location','best','FontSize',8);
grid on; xlim([0 A_ratio_exit+2]);
```

```
% 2 — Temperature vs Mach
subplot(2,3,2);
plot(M_sub, T_sub, 'b-', 'LineWidth',2); hold on;
plot(M_sup, T_sup, 'r-', 'LineWidth',2);
plot(M_t, T_t, 'ko','MarkerFaceColor','k','MarkerSize',8);
plot(M_e, T_e, 'rs','MarkerFaceColor','r','MarkerSize',8);
xlabel('Mach number M [-]'); ylabel('Temperature T [K]');
title('Temperature vs Mach number');
legend('Subsonic','Supersonic', ...
sprintf('Throat %.0fK',T_t), ...
sprintf('Exit %.0fK',T_e),'Location','best','FontSize',8);
grid on;
% 3 — Pressure vs Mach
subplot(2,3,3);
plot(M_sub, P_sub/1e6, 'b-','LineWidth',2); hold on;
plot(M_sup, P_sup/1e6, 'r-','LineWidth',2);
plot(M_t, P_t/1e6,'ko','MarkerFaceColor','k','MarkerSize',8);
plot(M_e, P_exit/1e6,'rs','MarkerFaceColor','r','MarkerSize',8);
xlabel('Mach number M [-]'); ylabel('Pressure P [MPa]');
title('Pressure vs Mach number');
legend('Subsonic','Supersonic', ...
sprintf('Throat %.3fMPa',P_t/1e6), ...
sprintf('Exit %.4fMPa',P_exit/1e6),'Location','best','FontSize',8);
grid on;
% 4 — Engine contour
subplot(2,3,4);
r_sub_flip = fliplr(sqrt(AR_sub * A_t / pi));
r_sup_line = sqrt(AR_sup * A_t / pi);
x_sub = linspace(0, L_c+L_conv, length(M_sub));
x_sup = linspace(L_c+L_conv, L_c+L_conv+L_n, length(M_sup));
x_all = [x_sub, x_sup(2:end)];
r_all = [r_sub_flip, r_sup_line(2:end)];
fill([x_all, fliplr(x_all)],[r_all, -fliplr(r_all)], ...
[0.85 0.92 1.0],'EdgeColor',[0.2 0.4 0.8],'LineWidth',1.5);
hold on; yline(0,'k--','LineWidth',0.8);
xline(L_c+L_conv,'k:','Throat','LabelHorizontalAlignment','left','FontSize',8);
xlabel('Axial position x [m]'); ylabel('Radius r [m]');
title('Engine contour (schematic)'); grid on; axis equal;
% 5 — Isp vs O/F sensitivity
subplot(2,3,5);
OF_range = linspace(cea_table(1,1), cea_table(end,1), 120);
Isp_range = zeros(size(OF_range));
Ve_range = zeros(size(OF_range));
for k = 1:length(OF_range)
Tc_k = interp1(cea_table(:,1), cea_table(:,2), OF_range(k), 'pchip');
gam_k = interp1(cea_table(:,1), cea_table(:,3), OF_range(k), 'pchip');
Mm_k = interp1(cea_table(:,1), cea_table(:,4), OF_range(k), 'pchip');
```

```
Rg_k = R_univ / Mm_k;
Cs_k = (1/gam_k)*sqrt(gam_k*Rg_k*Tc_k)*((gam_k+1)/2)^((gam_k+1)/(2*(gam_k-1)));
t1 = (2*gam_k^2/(gam_k-1))*(2/(gam_k+1))^((gam_k+1)/(gam_k-1));
t2 = 1-(P_exit/P_c)^((gam_k-1)/gam_k);
CF_k = sqrt(t1*t2);
Ve_range(k) = CF_k * Cs_k;
Isp_range(k) = Ve_range(k) / g0;
end
yyaxis left;
plot(OF_range, Ve_range,'b-','LineWidth',2);
ylabel('Ve [m/s]');
yyaxis right;
plot(OF_range, Isp_range,'r--','LineWidth',2);
ylabel('Isp [s]');
xline(OF,'k:',sprintf('O/F=%.1f',OF),'FontSize',9);
xlabel('O/F [-]');
title('Ve and Isp vs O/F (CEA data)');
grid on;
% 6 — Summary
subplot(2,3,6); axis off;
txt = {
'── SUMMARY ──────────────────────────────'
sprintf(' Thrust F = %.0f N', F)
sprintf(' Chamber P P_c = %.2f MPa', P_c/1e6)
sprintf(' O/F ratio = %.2f', OF)
''
'── COMBUSTION ───────────────────────────'
sprintf(' T_chamber = %.1f K', T_c)
sprintf(' Gamma γ = %.4f', gamma)
sprintf(' M_mol = %.3f kg/kmol', M_mol)
sprintf(' R_gas = %.2f J/kg/K', R_gas)
''
'── PERFORMANCE ──────────────────────────'
sprintf(' C* = %.2f m/s', C_star)
sprintf(' C_F = %.4f', C_F)
sprintf(' Ve (eff.) = %.2f m/s', Ve_eff)
sprintf(' Isp = %.2f s', Isp)
sprintf(' Mass flow mdot = %.4f kg/s', m_dot)
sprintf(' LOX flow mdot_ox = %.4f kg/s', m_dot_ox)
sprintf(' LH2 flow mdot_f = %.4f kg/s', m_dot_fuel)
''
'── THROAT ───────────────────────────────'
sprintf(' M_throat = %.4f', M_t)
sprintf(' T_throat = %.2f K', T_t)
sprintf(' P_throat = %.4f MPa', P_t/1e6)
sprintf(' D_throat = %.1f mm', D_t*1000)
''
```

```
'── EXIT ──────────────────────────────────'
sprintf(' M_exit = %.4f', M_e)
sprintf(' T_exit = %.2f K', T_e)
sprintf(' D_exit = %.1f mm', D_e*1000)
sprintf(' Area ratio eps = %.4f', A_ratio_exit)
''
'── CHAMBER ───────────────────────────────'
sprintf(' D_chamber = %.1f mm', D_c*1000)
sprintf(' Contraction ratio = %.2f', contraction_ratio)
sprintf(' Total length = %.1f mm', L_total*1000)
};
text(0.02, 0.99, txt, 'Units','normalized','VerticalAlignment','top', ...
'FontSize',8,'FontName','Courier New','Color',[0.08 0.08 0.08]);
sgtitle(sprintf('H_2/O_2 Rocket Engine | F=%.0fN | P_c=%.1fMPa | O/F=%.2f |
Isp=%.1fs', ...
F, P_c/1e6, OF, Isp), 'FontSize',12,'FontWeight','bold');
```